CSE STUDY HUB

Emerging Technology

UNIT 3 · IoT, Robotics & Cyber-Physical Systems

What this unit is about

This unit builds understanding of **IoT, Robotics & Cyber-Physical Systems**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Internet of Things (IoT): devices, sensors, actuators and connectivity

2. IoT architecture: perception, network, processing and application layers

3. Communication protocols and edge gateways at a conceptual level

4. Robotics: sensors, actuators, controller, end effector and feedback

5. Automation, industrial robots and collaborative robots

6. Cyber-Physical Systems (CPS): computation integrated with physical processes

7. Applications in smart homes, factories, healthcare and transport

8. Challenges: latency, interoperability, safety, privacy and security

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 3 NOTES · PART 1

Internet of Things (IoT): devices, sensors, actuators and connectivity

IoT systems connect physical devices containing sensors/actuators to networks and software. A typical flow is sensing → connectivity/gateway → processing → application/action; security and interoperability are key concerns.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

IoT architecture: perception, network, processing and application layers

IoT systems connect physical devices containing sensors/actuators to networks and software. A typical flow is sensing → connectivity/gateway → processing → application/action; security and interoperability are key concerns.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Communication protocols and edge gateways at a conceptual level

Communication protocols and edge gateways at a conceptual level is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on IoT, Robotics & Cyber-Physical Systems, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 2

Robotics: sensors, actuators, controller, end effector and feedback

Robotics: sensors, actuators, controller, end effector and feedback is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Automation, industrial robots and collaborative robots

Automation, industrial robots and collaborative robots is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Cyber-Physical Systems (CPS): computation integrated with physical processes

Present Cyber-Physical Systems (CPS): computation integrated with physical processes as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on IoT, Robotics & Cyber-Physical Systems, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 3

Applications in smart homes, factories, healthcare and transport

For Applications in smart homes, factories, healthcare and transport, connect the concept to a practical scenario. Identify the input or need, the technology/process used, the output or benefit, and one limitation. Strong examples explain not only what is used but why it is suitable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Challenges: latency, interoperability, safety, privacy and security

Compare the alternatives in Challenges: latency, interoperability, safety, privacy and security using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on IoT, Robotics & Cyber-Physical Systems, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Internet of Things (IoT): devices, sensors, actuators and connectivity** — know the definition, core working and one example.

• **IoT architecture: perception, network, processing and application layers** — know the definition, core working and one example.

• **Communication protocols and edge gateways at a conceptual level** — know the definition, core working and one example.

• **Robotics: sensors, actuators, controller, end effector and feedback** — know the definition, core working and one example.

• **Automation, industrial robots and collaborative robots** — know the definition, core working and one example.

• **Cyber-Physical Systems (CPS): computation integrated with physical processes** — know the definition, core working and one example.

• **Applications in smart homes, factories, healthcare and transport** — know the definition, core working and one example.

• **Challenges: latency, interoperability, safety, privacy and security** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from IoT, Robotics & Cyber-Physical Systems. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Internet of Things (IoT): devices, sensors, actuators and connectivity and explain its importance.

Q2. Explain the working/steps involved in IoT architecture: perception, network, processing and application layers.

Q3. Differentiate between two related concepts from IoT, Robotics & Cyber-Physical Systems.

Q4. Give one practical application of Challenges: latency, interoperability, safety, privacy and security and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of IoT, Robotics & Cyber-Physical Systems.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.